

# Affine-DDPNM: Methodology

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## 1 Introduction

We extend the domain-decomposition pore-network method (DD-PNM) of [1] by enriching the interface traction space from a single constant-normal mode to a nine-mode affine representation. The enriched method, termed **Affine-DDPNM**, retains the same offline–online workflow and non-overlapping subdomain partition of the original DD-PNM. The only change is the interface Ansatz: on each internal face, the traction is now allowed to vary as an affine function of the in-plane coordinates, in both the normal and the two tangential directions.

This section is structured as a self-contained methodology description. Readers familiar with the classic DD-PNM [1] may note that Sections 2–3 are identical to the original formulation; the enrichment appears in Sections 5–7.

## 2 Model problem

We consider steady, incompressible Stokes flow in a three-dimensional pore domain  $\Omega \subset \mathbb{R}^3$ . Let  $\Gamma_{\text{wall}}$  be the solid–fluid interface. A pressure drop  $p_{\text{in}} - p_{\text{out}} > 0$  is imposed between the inlet  $\Gamma_{\text{in}}$  and outlet  $\Gamma_{\text{out}}$  boundaries, while all remaining external boundaries are treated as no-slip walls. Collecting all homogeneous Dirichlet segments in  $\Gamma_{\text{dir}} := \Gamma_{\text{wall}} \cup \Gamma_{\text{top}} \cup \Gamma_{\text{bottom}} \cup \Gamma_{\text{front}} \cup \Gamma_{\text{back}}$ , the governing equations read

$$-\nu \Delta \mathbf{u} + \nabla p = \mathbf{0}, \quad \text{in } \Omega, \quad (1)$$

$$\nabla \cdot \mathbf{u} = 0, \quad \text{in } \Omega, \quad (2)$$

$$\mathbf{u} = \mathbf{0}, \quad \text{on } \Gamma_{\text{dir}}, \quad (3)$$

$$\mathbf{n} \cdot \boldsymbol{\sigma}(\mathbf{u}, p) \mathbf{n} = -p_b, \quad \mathbf{t} \cdot \boldsymbol{\sigma}(\mathbf{u}, p) \mathbf{n} = \mathbf{0}, \quad \text{on } \Gamma_{\text{in}} \cup \Gamma_{\text{out}}, \quad (4)$$

where  $\mathbf{u}$  is velocity,  $p$  is pressure,  $\nu$  is the kinematic viscosity,  $\boldsymbol{\sigma}(\mathbf{u}, p) := -p\mathbf{I} + \nu \nabla \mathbf{u}$  is the Cauchy stress tensor,  $\mathbf{n}$  is the outward unit normal, and  $p_b = p_{\text{in}}$  on  $\Gamma_{\text{in}}$ ,  $p_b = p_{\text{out}}$  on  $\Gamma_{\text{out}}$ .

## 3 Domain partition

The pore domain  $\Omega$  is partitioned into  $N$  non-overlapping Lipschitz subdomains  $\{\Omega_i\}_{i=1}^N$ . Each subdomain corresponds to one pore body. For any pair  $(i, j)$  of adjacent pores, the shared interface is  $\gamma_k = \Omega_i \cap \Omega_j$ ,  $k = 1, \dots, m$ , where  $m$  is the total number of internal interfaces. The interfaces are constructed by the Voronoi (or watershed) partition of the void space: each interface is the plane (or surface) equidistant from the two neighbouring solid grains.

For each subdomain  $\Omega_i$  we list its  $m_i$  ports as  $\{\gamma_{i,1}, \dots, \gamma_{i,m_i}\}$ , partitioned into an unknown-traction set  $\mathcal{U}_i$  (internal interfaces) and a known-traction set  $\mathcal{K}_i$  (inlet/outlet boundaries). The local-to-global index maps  $\iota_i^u$  and  $\iota_i^k$  and the associated Boolean restriction matrices  $R_i^u$ ,  $R_i^k$  are defined exactly as in [1, Section 2.4].

## 4 Reference finite-element discretization

Within each subdomain (and globally, for the reference solution) we employ the inf-sup stable Taylor–Hood pair  $\mathbf{V}_h = [P_2]^3 \cap \mathbf{V}$ ,  $Q_h = P_1 \cap Q$  on a body-fitted tetrahedral mesh. The velocity and pressure spaces are

$$\mathbf{V} = \{\mathbf{v} \in [H^1(\Omega)]^3 : \mathbf{v} = \mathbf{0} \text{ on } \Gamma_{\text{dir}}\}, \quad Q = L^2(\Omega).$$

With bases  $\{\phi_j\}_{j=1}^{n_u}$  and  $\{\psi_l\}_{l=1}^{n_p}$ , the discrete system on a subdomain  $\Omega_i$  reads

$$\begin{bmatrix} A_i & B_i^\top \\ B_i & 0 \end{bmatrix} \begin{bmatrix} \mathbf{U}_i \\ \mathbf{P}_i \end{bmatrix} = \sum_{r=1}^{m_i} \mathbf{f}_i^{(r)}, \quad (5)$$

where  $A_i$  and  $B_i$  are the local Stokes stiffness and divergence matrices, and  $\mathbf{f}_i^{(r)}$  is the Neumann load vector assembled from the traction prescribed on port  $r$ .

## 5 Interface traction Ansatz

### 5.1 Classic DD-PNM (one mode per interface)

In the classic DD-PNM [1], the traction on each internal interface  $\gamma$  is assumed to be uniform and purely normal:

$$\mathbf{n} \cdot \boldsymbol{\sigma}(\mathbf{u}, p) \mathbf{n} = -p_\gamma, \quad \mathbf{t} \cdot \boldsymbol{\sigma}(\mathbf{u}, p) \mathbf{n} = \mathbf{0}, \quad (6)$$

where  $p_\gamma \in \mathbb{R}$  is a single scalar unknown per interface. This corresponds to approximating the true traction trace by the **one-dimensional subspace**

$$\mathcal{T}_\gamma^{\text{P0}} = \text{span}\{\mathbf{n}\}.$$

### 5.2 Affine-DDPNM (nine modes per interface)

In the Affine-DDPNM, we enrich the interface traction space to allow affine spatial variation in both the normal and tangential components. For each interface  $\gamma$ , we introduce a local orthonormal frame  $\{\mathbf{n}_\gamma, \mathbf{t}_{\gamma,1}, \mathbf{t}_{\gamma,2}\}$  with origin at the interface centroid  $\mathbf{c}_\gamma$ . The in-plane coordinates are

$$s(\mathbf{x}) = \frac{(\mathbf{x} - \mathbf{c}_\gamma) \cdot \mathbf{t}_{\gamma,1}}{\ell_\gamma}, \quad t(\mathbf{x}) = \frac{(\mathbf{x} - \mathbf{c}_\gamma) \cdot \mathbf{t}_{\gamma,2}}{\ell_\gamma}, \quad (7)$$

where  $\ell_\gamma = \sqrt{|\gamma|}$  is a characteristic interface length scale that renders  $s, t$  dimensionless and  $O(1)$  on  $\gamma$ .

The traction on  $\gamma$  is then expanded as

$$\boldsymbol{\sigma}(\mathbf{u}, p) \mathbf{n}_\gamma = \sum_{\alpha \in \{0, s, t\}} \sum_{\beta \in \{\mathbf{n}, \mathbf{t}_1, \mathbf{t}_2\}} \lambda_{\gamma, \alpha, \beta} \underbrace{\varphi_\alpha(s, t)}_{\text{scalar shape}} \underbrace{\mathbf{d}_\beta}_{\text{direction}}, \quad (8)$$

where

$$\varphi_0(s, t) = 1, \quad \varphi_s(s, t) = s, \quad \varphi_t(s, t) = t,$$

and  $\mathbf{d}_\mathbf{n} = \mathbf{n}_\gamma$ ,  $\mathbf{d}_{\mathbf{t}_1} = \mathbf{t}_{\gamma,1}$ ,  $\mathbf{d}_{\mathbf{t}_2} = \mathbf{t}_{\gamma,2}$ .

The nine scalar coefficients  $\boldsymbol{\lambda}_\gamma = (\lambda_{\gamma,0,\mathbf{n}}, \lambda_{\gamma,0,\mathbf{t}_1}, \lambda_{\gamma,0,\mathbf{t}_2}, \lambda_{\gamma,s,\mathbf{n}}, \dots, \lambda_{\gamma,t,\mathbf{t}_2}) \in \mathbb{R}^9$  are the unknown interface degrees of freedom on  $\gamma$ .

**Remark 1.** *The nine-mode space can be written compactly as*

$$\mathcal{T}_\gamma^{\text{Affine}} = \text{span}\{\varphi_\alpha(s, t) \mathbf{d}_\beta \mid \alpha \in \{0, s, t\}, \beta \in \{\mathbf{n}, \mathbf{t}_1, \mathbf{t}_2\}\}.$$

*It contains the classic P0 normal mode ( $\alpha = 0, \beta = \mathbf{n}$ ) as a subspace. The enrichment adds (i) linear normal variation ( $s\mathbf{n}, t\mathbf{n}$ ), capturing non-uniform pressure drops across the interface, and (ii) constant and linear tangential tractions, capturing cross-flow and shear at pore throats.*

**Remark 2** (Sign convention). *The outward normal  $\mathbf{n}_\gamma$  changes sign depending on which pore is under consideration. For pore  $\Omega_i$  adjacent to interface  $\gamma$ , we define the side sign  $\sigma_{i,\gamma} = +1$  if  $\Omega_i$  is the first member of the interface pair and  $\sigma_{i,\gamma} = -1$  otherwise. The tangential directions are multiplied by  $\sigma_{i,\gamma}$  so that the flux continuity condition  $Q_{i,\gamma} + Q_{j,\gamma} = 0$  remains consistent under the sign flip of the normal.*

## 6 Local Dirichlet-to-Neumann map

### 6.1 Primitive response loads

For each subdomain  $\Omega_i$  and each of its  $m_i$  ports, we construct 9 primitive Neumann loads (or 1 for inlet/outlet ports, which retain the classic uniform-normal treatment). On an internal interface  $\gamma$ , the 9 unit-load problems are

$$\mathbf{f}_{i,\gamma}^{(\alpha,\beta)} \in \mathbb{R}^{n_{i,u}}, \quad [\mathbf{f}_{i,\gamma}^{(\alpha,\beta)}]_j = - \int_\gamma \varphi_\alpha(s, t) \mathbf{d}_\beta \cdot \boldsymbol{\phi}_j \, dS, \quad (9)$$

for  $\alpha \in \{0, s, t\}$ ,  $\beta \in \{\mathbf{n}, \mathbf{t}_1, \mathbf{t}_2\}$ .

Solving the local Stokes system for each unit load yields the velocity response  $\mathbf{U}_{i,\gamma}^{(\alpha,\beta)}$  and the pressure response  $\mathbf{X}_{i,\gamma}^{(\alpha,\beta)}$ :

$$\begin{bmatrix} A_i & B_i^\top \\ B_i & 0 \end{bmatrix} \begin{bmatrix} \mathbf{U}_{i,\gamma}^{(\alpha,\beta)} \\ \mathbf{X}_{i,\gamma}^{(\alpha,\beta)} \end{bmatrix} = \begin{bmatrix} \mathbf{f}_{i,\gamma}^{(\alpha,\beta)} \\ \mathbf{0} \end{bmatrix}. \quad (10)$$

### 6.2 Assembled DtN operator

Let the local traction vector on pore  $i$  collect all  $n_i = 9m_i^u + m_i^k$  unknowns (internal interfaces use 9 modes each; boundary ports use 1):

$$\boldsymbol{\lambda}_i \in \mathbb{R}^{n_i}, \quad n_i = 9m_i^u + m_i^k.$$

By linearity of (5), the discrete velocity is

$$\mathbf{U}_i = \sum_{\gamma \in \mathcal{U}_i} \sum_{\alpha, \beta} \lambda_{i,\gamma,\alpha,\beta} \mathbf{U}_{i,\gamma}^{(\alpha,\beta)} + \sum_{\gamma \in \mathcal{K}_i} \lambda_{i,\gamma} \mathbf{U}_{i,\gamma}^{(0,\mathbf{n})}.$$

The outward flux through port  $\gamma$  in mode  $(\alpha', \beta')$  is

$$Q_{i,\gamma}^{(\alpha',\beta')} = \int_\gamma \mathbf{u}_{i,h} \cdot (\sigma_{i,\gamma} \mathbf{n}_\gamma) \varphi_{\alpha'}(s, t) \, dS \quad (11)$$

and the total flux through  $\gamma$  (scalar, used for continuity) is the constant-mode component  $Q_{i,\gamma}^{(0,\mathbf{n})}$ .

The local DtN (Dirichlet-to-Neumann, or more precisely traction-to-flux) map  $\mathbf{G}_i \in \mathbb{R}^{n_i \times n_i}$  is assembled from the flux responses:

$$\mathbf{G}_i[(\gamma, \alpha, \beta), (\gamma', \alpha', \beta')] = -[\mathbf{f}_{i,\gamma}^{(\alpha,\beta)}]^\top \mathbf{U}_{i,\gamma'}^{(\alpha',\beta')}. \quad (12)$$

For boundary ports, only the  $(\alpha, \beta) = (0, \mathbf{n})$  entry is stored. As in the classic case,  $\mathbf{G}_i$  is symmetric negative semi-definite with the uniform-traction vector in its kernel [1].

## 7 Global Schur complement assembly

Let  $\boldsymbol{\lambda} \in \mathbb{R}^{9m}$  collect all unknown interface coefficients and  $\boldsymbol{\lambda}^k \in \mathbb{R}^{m^k}$  the prescribed inlet/outlet tractions. The restriction matrices  $R_i^u, R_i^k$  are block-structured to handle the nine-mode grouping: for each internal interface  $\gamma$ , the  $9 \times 9$  block of  $\mathbf{G}_i$  is assembled into the global Schur complement  $S \in \mathbb{R}^{9m \times 9m}$ .

Enforcing mass flux continuity on every internal interface (constant-mode component only) yields

$$S \boldsymbol{\lambda} = \mathbf{F}, \quad (13)$$

where

$$S = - \sum_{i=1}^N (R_i^u)^\top \mathbf{G}_i^{uu} R_i^u, \quad \mathbf{F} = \sum_{i=1}^N (R_i^u)^\top \mathbf{G}_i^{uk} R_i^k \boldsymbol{\lambda}^k,$$

and  $\mathbf{G}_i^{uu}, \mathbf{G}_i^{uk}$  are the blocks of  $\mathbf{G}_i$  corresponding to unknown–unknown and unknown–known couplings respectively.

**Remark 3** (Size of the global system). *In the classic DD-PNM, the Schur system is  $m \times m$  (one unknown per interface). In the Affine-DDPNM, it is  $9m \times 9m$  (nine unknowns per interface). The system remains sparse and symmetric positive-definite after the sign flip, and its size is still proportional to the number of interfaces rather than the number of FE nodes.*

## 8 Solution algorithm

Algorithm ?? summarises the Affine-DDPNM workflow. The only differences from the classic algorithm are (i) the generation of 9 primitive loads per internal interface instead of 1, and (ii) the nine-fold increase in the size of the local DtN blocks and the global Schur system. The offline–online decomposition and the embarrassingly parallel structure are preserved unchanged.

**Algorithm 1:** Affine-DDPNM solver (assemble – couple – recover)

**Input:** subdomain meshes, FE pairs  $(V_{i,h}, Q_{i,h})$ , prescribed tractions  $\boldsymbol{\lambda}^k$ .

**Output:** unknown interface coefficients  $\boldsymbol{\lambda}$ , interface fluxes, and (optionally) local fields.

**1. Offline library** — For each pore body  $i = 1, \dots, N$  (in parallel):

- (a) Assemble local operators  $A_i, B_i$ .
- (b) For each internal interface  $\gamma \in \mathcal{U}_i$  and each mode  $(\alpha, \beta) \in \{0, s, t\} \times \{\mathbf{n}, \mathbf{t}_1, \mathbf{t}_2\}$ :
  - i. Assemble unit load  $\mathbf{f}_{i,\gamma}^{(\alpha,\beta)}$  via (9).
  - ii. Solve local Stokes system (10) for  $(\mathbf{U}_{i,\gamma}^{(\alpha,\beta)}, \mathbf{X}_{i,\gamma}^{(\alpha,\beta)})$ .
- (c) For each boundary port  $\gamma \in \mathcal{K}_i$ : assemble and solve the single uniform-normal load  $(\alpha, \beta) = (0, \mathbf{n})$ .
- (d) Form local DtN block  $\mathbf{G}_i$  via (12).
- (e) Accumulate  $S \leftarrow (R_i^u)^\top \mathbf{G}_i^{uu} R_i^u, \mathbf{F} \leftarrow (R_i^u)^\top \mathbf{G}_i^{uk} R_i^k \boldsymbol{\lambda}^k$ .

**2. Global solve** — Solve global Schur system  $S \boldsymbol{\lambda} = \mathbf{F}$  by sparse direct solver.

**3. Reconstruction** (optional, in parallel):

$$\mathbf{U}_i = \sum_{\gamma, \alpha, \beta} \lambda_{i,\gamma,\alpha,\beta} \mathbf{U}_{i,\gamma}^{(\alpha,\beta)}, \quad \mathbf{P}_i = \sum_{\gamma, \alpha, \beta} \lambda_{i,\gamma,\alpha,\beta} \mathbf{X}_{i,\gamma}^{(\alpha,\beta)}.$$

## 9 Classic as a special case

Setting  $\lambda_{\gamma,\alpha,\beta} = 0$  for all modes except  $(\alpha, \beta) = (0, \mathbf{n})$  recovers the classic DD-PNM formulation exactly. The Affine-DDPNM is therefore a strict superset: its solution space contains the classic

solution, and the additional eight degrees of freedom per interface can only improve (or leave unchanged) the representation of the true interface traction trace.

In the numerical experiments of the companion paper, the Affine-DDPNM reduces the velocity  $L^2$  error by 73–90% relative to the classic DD-PNM across three distinct 3D geometries (regular sphere packing, random sphere packing, and a dense 100-sphere benchmark), with a 23–51% increase in total computational cost.

## 10 Computational cost

For a pore subdomain  $\Omega_i$  with  $m_i$  ports, the local DtN matrix  $\mathbf{G}_i$  has size  $n_i \times n_i$  where  $n_i \approx 9m_i$  (versus  $m_i$  in the classic method). Building  $\mathbf{G}_i$  requires  $n_i$  independent local Stokes solves; this offline work is embarrassingly parallel across subdomains. The global Schur system  $S$  has size  $9m \times 9m$  (versus  $m \times m$  in the classic method) and is solved by a sparse direct solver.

The offline cost grows by a factor of approximately  $9\times$  in the number of local solves (since each interface generates 9 loads instead of 1), but the online cost remains dominated by the sparse LU factorisation of  $S$ , which scales as  $O((9m)^3)$  in the worst case but is extremely sparse in practice ( $S$  has at most  $9 \times (\text{max coordination}) \times 9$  non-zero blocks per row).

## References

- [1] S. Sun, Z. Wang, L. Zhang, J. Zhao. A domain decomposition approach to pore-network modeling of porous media flow. *arXiv:2510.13429v2*, 2026.
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- [3] M.J. Blunt. *Multiphase Flow in Permeable Media*. Cambridge University Press, 2017.